

Assignment-1

classmate

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Name - Kshitiz Reddy M

Reg No - A26MTA10008

Sub - Advanced Machine Learning

Q1) Conceptual Understanding

Explain the fundamental difference between simple linear regression and multiple linear regression. Provide an example scenario where each would be appropriate and justify your choice.

A) Simple linear regression models the relationship between one independent variable and one dependent variable: $y = \theta_0 + \theta_1 x$.

Multiple linear regression extends this to several independent variables: $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$

Simple regression example: Predicting a person weight from their height alone - a single, dominant predictor is enough.

Multiple regression example: Predicting house price from square footage, number of bedrooms, and age - price genuinely depends on several interacting factors, so a single - variable model would miss most of the signal.

Q2) Cost Function Analysis

Given the following dataset:

$$x: [1, 2, 3, 4, 5], y: [2, 4, 5, 4, 6]$$

Calculate the Mean Squared Error (MSE) for the hypothesis $h(x) = 1 + 0.8x$. Show all steps of your calculation.

A)	x	y	$h(x) = 1 + 0.8x$	$y - h(x)$	$(y - h(x))^2$
	1	2	$1.8 = 1 + 0.8$	0.2	0.04
	2	4	$2.6 = 1 + 1.6$	1.4	1.96
	3	5	$3.4 = 1 + 2.4$	1.6	2.56
	4	4	$4.2 = 1 + 3.2$	-0.2	0.04
	5	6	$5.0 = 1 + 4.0$	1.0	1.00

$$\begin{aligned}\text{Sum of squared errors} &= 0.04 + 1.96 + 2.56 + 0.04 \\ &\quad + 1.00 \\ &= 5.6\end{aligned}$$

$$MSE = \frac{1}{n} \sum (y - h(x))^2 \Rightarrow \frac{1}{5} \times 5.6 = \frac{5.6}{5}$$

$$= 1.12$$

Q3) Gradient Descent Application

For the simple linear regression model $h(x) = \theta_0 + \theta_1 x$ with initial parameters $\theta_0 = 0, \theta_1 = 0$, and learning rate $\alpha = 0.01$, perform one iteration of gradient descent using the data:

$$x: [2, 4], y: [3, 7]$$

Show all calculations including partial derivatives and parameter updates

A) $h(x) = \theta_0 + \theta_1 x$ $\theta_0 = 0, \theta_1 = 0, \alpha = 0.01, x = [2, 4], y = [3, 7], m = 2$

Predictions: $h(2) = 0, h(4) = 0$

Errors: $(h(x) - y) : -3, -7$

$$\text{Cost: } J(\theta) = \frac{1}{2m} \sum (h(x) - y)^2$$

Partial derivatives:

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{m} \sum (h(x) - y) = \frac{1}{2} (-3 + -7) = \frac{1}{2} (-10) = -5$$

$$\begin{aligned} \frac{\partial J}{\partial \theta_1} &= \frac{1}{m} \sum (h(x) - y) \cdot x = \frac{1}{2} (-3 + -7) \cdot (2, 4) \\ &= \frac{1}{2} \cdot (-6 - 28) = \frac{1}{2} \cdot -34 = -17 \end{aligned}$$

Updates:

$$\theta_{0, \text{new}} = \theta_0 - \alpha \cdot (-5) = 0 - 0.01(-5) = 0.05$$

$$\theta_{1, \text{new}} = \theta_1 - \alpha \cdot (-17) = 0 - 0.01(-17) = 0.17$$

Q4) Multiple Regression Interpretation

A real estate company developed a multiple regression model to predict house prices:

$$\text{Price} = 50,000 + 120 \times \text{Square Footage} + 15,000 \times \text{Bedrooms} - 5,000 \times \text{Age}$$

Interpret the coefficients and explain what this model suggest about:

- The relationship between square footage & price
- The value of additional bedrooms
- How age affects property value

A) • Square footage: Holding other variables fixed, each additional square foot adds 120 to predicted price - a strong positive, continuous relationship

• Bedrooms: Each additional bedroom adds 15,000 to price, holding square footage and age constant - this captures value from bedroom count independent of total size

• Age: Each additional year of age reduces price by 5,000 - older homes are worth less, all else equal, likely reflecting wear, outdated systems, or style

• Intercept (50,000): The theoretical baseline price when square footage, bedrooms, and age are all zero - not meaningful on its own, just a mathematical anchor.

Q5) Polynomial Regression Design

You're analyzing the relationship between temperature (x) and electricity consumption (y). Explain why polynomial regression might be more appropriate than simple linear regression for this scenario. What degree polynomial would you start with and why?

A) Temperature vs. electricity consumption is not monotonic - consumption is low at mild temperatures and rises at both cold extremes (heating) and hot extremes (cooling), producing a U- or a J-shaped curve. A straight line can't capture that: you need curvature.

Starting degree: 2 (quadratic) - $h(x) = \theta_0 + \theta_1 x + \theta_2 x^2$. This is the simplest model that can represent a U-shape (a single minimum/maximum). You'd only go higher (degree 3+) if residual analysis showed the quadratic still missed structure - and higher degrees risk overfitting on top of losing interpretability.

Q6) Matrix Formulation

Given the multiple regression scenario with two features:

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 1 & 1 & 5 \end{bmatrix}, \quad Y = \begin{bmatrix} 4 \\ 7 \\ 6 \end{bmatrix}$$

Write the normal equations in matrix form and explain how you would solve for the parameters θ .

A) The normal equations come from minimizing $J(\theta) = (1/2m) \|X\theta - y\|^2$, which gives:

$$X^T X \theta = X^T y$$

Solving for θ :

$$\theta = (X^T X)^{-1} X^T y$$

Process:

- Compute $X^T X$
- Compute $X^T y$
- Invert $(X^T X)$ - this only works if $X^T X$ is non-singular
- Multiply $(X^T X)^{-1}$ by $X^T y$ to get $\theta = [\theta_0, \theta_1, \theta_2]^T$ directly, without iterating

Q1) Practical Implementation

Describe the step-by-step process you would follow to implement polynomial regression for a dataset with one input feature. Include:

- Feature transformation
- Parameter initialization
- Gradient descent implementation
- Convergence criteria

- A)
- **Feature transformation:** Convert the single feature x into polynomial terms: $x, x^2, x^3, \dots, x^d$, producing a new design matrix. Also standardize/normalize each column since higher powers create widely different scales.
 - **Parameter initialization:** Initialize $\theta_0, \theta_1, \dots, \theta_d$ - typically to zeros or small random values.
 - **Gradient descent implementation:** For each iteration compute predictions $h(x) = \theta_0 + \theta_1 x + \dots + \theta_d x^d$, compute the cost $J(\theta)$, compute partial derivatives $\partial J / \partial \theta_j$ for each parameter, and simultaneously update all θ_j :
$$\theta_j := \theta_j - \alpha \cdot \partial J / \partial \theta_j$$
 - **Convergence criteria:** Stop when the changes in $J(\theta)$ between iterations falls below a small threshold, or when a maximum iteration count is reached, or when the gradient magnitude is near zero. Monitoring $J(\theta)$ vs. iteration should show a smooth decrease - oscillation signals too high a learning rate.

Q8) Model Evaluation

A polynomial regression model of degree 3 achieved the following results:

- Training MSE: 2.5
- Validation MSE: 15.8
- Testing MSE: 16.2

Analyze these results and answer:

- What phenomenon is occurring?
- What would you do to improve the model?
- What degree polynomial might be more appropriate?

- A)
- What's happening: This is overfitting. The large gap between training error and validation / testing error means the degree-3 model has fit noise and idiosyncrasies of training set rather than the true underlying pattern, so it doesn't generalize.
 - How to improve it:
 - Reduce model complexity (lower polynomial degree)
 - Add regularization (Ridge / L2 or Lasso / L1) to penalize large coefficients
 - Get more training data
 - Use cross-validation to tune degree / regularization strength systematically
 - More appropriate degree: Likely degree 1 or 2 - start simpler and only increase degree if validation error also improves, not just training error.

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Q9) Numerical Calculation

For the quadratic regression model $h(x) = \theta_0 + \theta_1 x + \theta_2 x^2$ with:

$$\theta_0 = 1, \theta_1 = 0.5, \theta_2 = 0.2$$

Calculate the predicted values for $x = [1, 2, 3]$ and compute the MSE if the actual y values are $[2, 4, 8]$

A) $h(x) = 1 + 0.5x + 0.2x^2$; $y = [2, 4, 8]$, $x = [1, 2, 3]$

- $h(1) = 1 + 0.5(1) + 0.2(1)^2 = 1 + 0.5 + 0.2 = 1.7$
- $h(2) = 1 + 0.5(2) + 0.2(2)^2 = 1 + 1.0 + 0.8 = 2.8$
- $h(3) = 1 + 0.5(3) + 0.2(3)^2 = 1 + 1.5 + 1.8 = 4.3$

x	y	$h(x)$	$y - h(x)$	$(y - h(x))^2$
1	2	1.7	0.3	0.09
2	4	2.8	1.2	1.44
3	8	4.3	3.7	13.69

$$\text{Sum} = 0.09 + 1.44 + 13.69 = 15.22$$

$$\text{MSE} = 15.22 / 3 = \approx 5.073$$

Q10) Real-world Application

Design a multiple regression model to predict Student academic performance. Identify:

- 3-4 relevant features you would include
- The expected sign of each coefficient (+ or -)
- One potential interaction term you might consider
- How you would validate your model's effectiveness

A) Features:

- Attendance rate (%) - expected sign: + (more attendance better performance)
- Hours studies per week - expected sign: +
- Prior GPA / prior test scores - expected sign: + (Strong predictor of future performance)
- Hours of part-time work per week - expected sign: - (Less time for study, more fatigue)

Interaction terms: Study hours x Prior GPA - a student with a strong academic foundations may benefit more from each additional study hour than a struggling student, so this interaction captures that the effect of studying isn't uniform across ability levels

Validation approach: Split data into training/validation/test sets, evaluate using MSE/ R^2 on held-out data, check residual plots for patterns and compare against a baseline to confirm the model adds real predictive value.

8) Bonus : Optimization Challenge

Given the cost function $J(\theta) = 1/6 (\theta_0 + \theta_1 x - y)^2$ for three data points, derive the gradient descent update rules for both θ_0 and θ_1 . Explain how the learning rate affects convergence and what happens if it's set too high or too low.

A) $J(\theta) = (1/6) \sum (\theta_0 + \theta_1 x - y)^2$ for three data points - note $1/6 = 1/(2m)$ with $m=3$, so this is the standard MSE cost form

Gradients:

$$\partial J / \partial \theta_0 = (1/3) \sum (\theta_0 + \theta_1 x - y)$$

$$\partial J / \partial \theta_1 = (1/3) \sum (\theta_0 + \theta_1 x - y) \cdot x$$

Update rules:

$$\theta_0 := \theta_0 - \alpha \cdot (1/3) \sum (\theta_0 + \theta_1 x - y)$$

$$\theta_1 := \theta_1 - \alpha \cdot (1/3) \sum (\theta_0 + \theta_1 x - y) x$$

Effect of Learning rate α :

- Too high: Updates overshoots the minimum; $J(\theta)$ can oscillate or even diverge instead of converging.
- Too low: Convergence is very slow, requiring many iterations and more computations to reach the minimum, though it will still converge steadily in the right direction.
- Well-chosen α : $J(\theta)$ decreases smoothly and reaches a minimum in a reasonable number of iterations.